

Theory-Explicit Prompting for MIND Self-States: Hierarchical LLMs and Dynamic Signature Extraction in Mental Health Timelines

First Author	Second Author	Third Author	Fourth Author
CSE, IIT Ropar	CSE, IIT Ropar	CSE, IIT Roapr	CSE, IIT Ropar
email@iitrpr.ac.in	email@iitrpr.ac.in	email@iitrpr.ac.in	email@iitrpr.ac.in

Abstract

This paper presents a system for the CLPsych 2026 Shared Task on longitudinal mental health modeling from social media timelines, grounded in the MIND framework (Atzil-Slonim, 2025). MIND conceptualizes mental health as evolving self-states defined by Affect, Behavior, Cognition, and Desire (ABCD), providing a structured lens on mental health trajectories. The system centers on a *theory-explicit prompting framework* for structured sequence summarization (Task 3.1) and recurrent dynamic signature extraction (Task 3.2), encoding the full ABCD taxonomy directly into the LLM prompt to ensure clinically grounded, interpretable outputs. A three-stage pipeline infers a direction-of-change label per sequence, produces structured ABCD summaries with few-shot exemplar augmentation, and aggregates these summaries to derive cross-individual recurrent patterns. The system ranks *first* on deterioration-related recurrent signatures and *second* overall, achieving the top Fit and Specificity scores in Task 3.2, demonstrating the benefits of explicit clinical grounding for conceptual accuracy.

1 Introduction

Social media platforms, particularly Reddit, offer rich longitudinal records of self-disclosure that unfold over weeks, months, and years, enabling forms of continuous monitoring and early risk detection that single-point clinical assessments cannot easily provide (Coppersmith et al., 2014; Tsakalidis and Liakata, 2022). As a result, these platforms have become a key resource for computational mental health, supporting work on early warning signals, symptom trajectories, and crisis detection.

Prior work has examined diverse aspects of mental health-relevant language in user-generated content, including how linguistic and contextual properties vary across societal and non-societal topics

(Singh and Singh, 2024), the extraction and summarization of suicidal ideation using large language models (Singh et al., 2024a), and the detection of critical moments of change through longitudinal and multi-task learning approaches (Azim et al., 2022; Singh et al., 2024b). While these methods demonstrate strong predictive performance, they typically target isolated subtasks and often remain weakly connected to established psychological theory, limiting interpretability, complicating clinical integration, and constraining our ability to model mental-state evolution in ways that are meaningful to clinicians.

The CLPsych 2026-25 Shared Task (Ali et al., 2026)(Tseriotou et al., 2025) addresses this gap by placing longitudinal mental health modeling within the MIND framework (Atzil-Slonim, 2025), which conceptualizes mental health as a dynamic configuration of *self-states* shaped by Affect (A), Behavior toward the self (B-S) and toward others (B-O), Cognition about the self (C-S) and about others (C-O), and Desire (D)—collectively referred to as ABCD. At any point in time, one self-state configuration may be dominant while others remain latent, and shifts between more adaptive and more maladaptive self-states are intended to reflect clinically meaningful patterns of deterioration or recovery. The task spans 40 Reddit-based user timelines from mental health subreddits, split into 30 training and 10 test timelines (Ali et al., 2026), each consisting of chronologically ordered posts annotated according to the MIND scheme.

This paper concentrates on the sequence-level tasks of the CLPsych 2026 Shared Task (Task 3). Details on the task and the dataset are provided in Appendix A. Task 3.1 requires generating structured natural-language summaries of the ABCD dynamics surrounding each detected change event (Switch or Escalation), while Task 3.2 builds on these summaries to identify *recurrent* ABCD-grounded dynamic signatures that distinguish de-

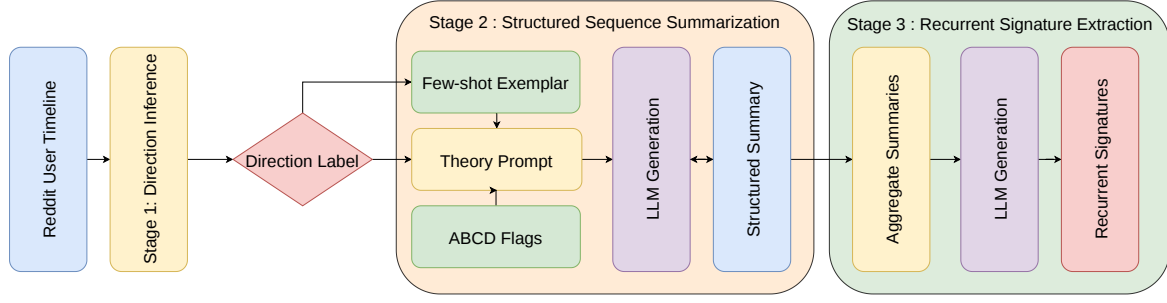


Figure 1: Three-stage pipeline for Task 3. Purple nodes denote LLM generation steps; yellow nodes denote prompt construction and aggregation; green nodes denote external context inputs. Stage 1 infers a direction-of-change label from each post sequence. Stage 2 assembles a theory-explicit MIND/ABCD prompt with a matched few-shot exemplar and per-post annotations, and generates a structured summary (Task 3.1). Stage 3 aggregates training summaries by direction and extracts recurrent dynamic signatures across users (Task 3.2).

terioration from improvement across multiple individuals’ trajectories. We address these tasks through a *theory-explicit prompting framework* that encodes the complete MIND/ABCD taxonomy directly into the system prompt to guide structured sequence summarization and signature extraction. Each summary is organized into three clinically motivated sections: *Central Theme*, *Within-State Dynamics*, and *Between-State Dynamics*, and uses formal ABCD abbreviations and relational vocabulary to support theory-aligned, clinically interpretable modeling of longitudinal mental health.

Building on this design, a three-stage pipeline drives the system: (1) a keyword-based heuristic infers a direction-of-change label (*deterioration*, *improvement*, or *mixed*) for each sequence; (2) this label conditions a two-part structured prompt, with a matched few-shot exemplar retrieved from training, that produces the Task 3.1 ABCD summary; and (3) these summaries are aggregated across users to perform cross-individual signature extraction in Task 3.2. Our pipeline achieves *first place* in identifying deterioration signatures and *second place* overall in Task 3.2, attaining the highest Fit Score (1.000 overall; 0.875 deterioration) and Specificity Score (0.938) of any participating team in the deterioration category. These results illustrate how theory-explicit prompting can bridge the gap between high-performing LLMs and psychologically interpretable characterizations of mental health trajectories.

2 Methodology: Summarization and Signature Extraction

Our sequence-level component uses a unified theory-explicit prompt-based LLM pipeline

to address structured sequence summarization (Task 3.1) and recurrent pattern extraction (Task 3.2). We encode the ABCD taxonomy, self-state definitions, and the required output format directly in the system prompt so that all generations are explicitly grounded in the MIND framework (Atzil-Slonim, 2025).

The pipeline has three stages: (1) infer a direction-of-change label for each sequence; (2) generate a structured summary for Task 3.1 conditioned on this label; and (3) aggregate Task 3.1 summaries over the training set to extract cross-individual signatures for Task 3.2. The full pipeline is illustrated in Figure 1.

2.1 Stage 1: Direction Inference

Each sequence is labeled as *deterioration*, *improvement*, or *mixed* by scanning its gold summary (training) for deterioration keywords (e.g., *worsen*, *suicidal*) and improvement keywords (e.g., *recover*, *hopeful*). The higher count determines the label; ties yield *mixed*. For test sequences, the same heuristic is applied to the post texts. This direction labels both frames, the kind of change described in Task 3.1, and groups Task 3.1 summaries when building signatures in Task 3.2.

2.2 Stage 2: Task 3.1 – Structured Sequence Summarization

Given a direction label, we construct a two-part prompt: a fixed system prompt and a sequence-specific user prompt. The system prompt asks the model to act as a clinician trained in MIND and to produce a three-part summary:

- **Central Theme:** the dominant ABCD dimension driving the change and how it evolves

across the sequence.

- **Within-State Dynamics:** interactions *within* self-states (e.g., co-activation, reinforcement, suppression) between ABCD subelements.
- **Between-State Dynamics:** how adaptive and maladaptive self-states change in relative dominance, culminating in a Switch or unfolding through an Escalation.

The model must use formal third-person clinical prose, explicitly reference ABCD elements with (A), (B-S), (B-O), (C-S), (C-O), (D), state the direction of change, and not reproduce numeric scores. Outputs are capped at 350 words to match the shared task truncation limit.

The user prompt provides sequence identifiers and change types (Switch / Escalation), the inferred direction label, and for each post: its index, inferred self-state valence (adaptive / maladaptive / neutral), ABCD presence flags with scores, and post text truncated to 500 characters. These per-post flags anchor the model in the structured annotations and reduce hallucinated ABCD labels.

Few-Shot Augmentation. We add a few-shot exemplar summary to the user prompt. For a sequence with change type c and direction d , we retrieve a training summary whose (change type, direction) best matches (c, d) (exact match; otherwise, direction-only; otherwise, most recent). It is inserted under EXAMPLE SUMMARY (for style reference only), followed by Now generate a summary for the following new sequence, preventing contamination of the target input.

The exemplar acts as a stylistic and structural template: it encourages the correct use of ABCD abbreviations, relational language (e.g., *mutually reinforces, progressively suppresses*), and the three-part structure. Decoding uses temperature $\tau = 0.3$ with a 700-token limit. Outputs are split into three sections via regex over numbered headers; if parsing fails, the full output is kept as a single summary.

2.3 Stage 3: Task 3.2 – Recurrent Dynamic Signature Extraction

For Task 3.2, we use the training-side Task 3.1 summaries as input to extract recurrent ABCD-based signatures across users. Summaries are grouped by direction label (*deterioration, improvement, mixed*). A single aggregated prompt then presents all

Team	CS $\uparrow$	CT $\downarrow$	R-L $\uparrow$	BS $\uparrow$	Rank
MERONYM_LABS	0.801	0.659	0.266	0.345	1
DreamerNLplus	0.735	0.767	0.285	0.345	2
CUNY	0.789	0.714	0.292	0.295	3
Ours	0.688	0.812	0.242	0.306	11
Baseline	0.765	0.749	0.262	0.231	–

Table 1: Results of Task 3.1 (selected top 3 teams vs. ours). CS = Consistency; CT = Contradiction; R-L = ROUGE-L Recall; BS = BERTScore Recall. Bold = best per column.

summaries, organized by group and truncated to 800 characters each.

The model is instructed to report only signatures that occur in *at least two distinct user timelines*. For each signature, it must specify: (i) the ABCD interaction pattern (which subelements co-activate or suppress adaptive functioning); (ii) how adaptive vs. maladaptive dominance evolves before the change event; (iii) how the dynamics crystallize at the Switch or Escalation; and (iv) two to three training timeline IDs as evidence.

Signature extraction uses temperature $\tau = 0.2$ with up to 1,500 new tokens, favoring stable, high-level patterns. We parse outputs by splitting on deterioration signature and improvement signatures, then apply field-specific regex to extract each component. When parsing fails, we retain the raw output for manual inspection.

3 Results and Analysis

We focus our analysis on Task 3, where our system is most closely aligned with the MIND framework and achieves its strongest performance. Quantitative results are reported on the 10 test timelines provided by the shared task organizers, and we complement these with a qualitative analysis of the extracted deterioration signatures.

3.1 Task 3.1: Summarisation Quality

Table 1 reports the official Task 3.1 evaluation across four metrics: Consistency (CS)—absence of contradiction against gold summaries via NLI; Contradiction (CT)—maximum NLI contradiction probability (lower is better); ROUGE-L Recall—longest common subsequence recall; and BERTScore Recall—semantic token-level coverage. The final rank is based on the average score rank across all four metrics. Our system (rank 11) shows moderate CS (0.688) and BERTScore (0.306) with a relatively high CT (0.812), indi-

Team	Fit	Recur	Spec	Overall
Task 3.2 Deterioration extraction results				
Ours	0.875	0.563	0.938	0.789
Aurevia	0.688	0.813	0.563	0.676
DreamerNLplus	0.438	0.688	0.875	0.604
MeronymLabs	0.438	0.563	0.813	0.551
psytechlab	0.625	0.625	0.250	0.491
Baseline	0.563	0.375	0.438	0.483
Overall Task 3.2 competition results				
Ours	1.000	0.688	0.375	0.743
DreamerNLplus	0.625	0.813	1.000	0.761
MKC	0.750	0.563	0.938	0.727
McMasterNLP	0.688	0.375	0.750	0.594
psytechlab	0.688	1.000	0.250	0.544
Aurevia	0.375	0.625	0.500	0.465
Baseline	0.375	0.438	0.375	0.389

Table 2: Results for *Improvement* and *Deterioration*. Systems are ranked by *Overall*, which aggregates *Fit* (alignment with the ABCD framework), *Recur* (cross-user recurrence), and *Spec* (specificity vs. genericness). Our system ranks *first* in *Deterioration* and *second* in *Improvement*, with the highest *Fit* in both.

cating that our ABCD-grounded summaries are internally coherent but diverge in phrasing from the gold annotations. The high CT score suggests that our summaries occasionally emphasize different aspects of the ABCD dynamics than the gold, even though they cover the same clinical content. Systems with high ROUGE-L (e.g., USAI: 0.333) likely produce summaries with greater lexical overlap with the gold text, whereas our approach prioritizes clinical framework fidelity over lexical reproduction—a trade-off also reflected in our strong Task 3.2 Fit Score.

3.2 Task 3.2: Quantitative Performance

Tables 2 report the official Task 3.2 results. Scores assess: *Fit* (alignment with the ABCD clinical framework), *Recurrence* (cross-individual pattern grounding), and *Specificity* (non-genericity of the description).

MIND Framework Alignment Our perfect Fit Score (1.000 overall; 0.875 deterioration) reflects the effectiveness of theory-explicit prompting: encoding the ABCD taxonomy directly into the system prompt ensures that signatures closely match the clinical framework used for evaluation, outperforming all other teams in this dimension.

Fit-Specificity Trade-off Our lower Specificity Score in the overall ranking (0.375) reflects a trade-off inherent to our aggregation strategy. Presenting

all training summaries simultaneously encourages cross-individual abstraction at the cost of per-case granularity. This contrasts with our high deterioration Specificity (0.938), suggesting that deterioration patterns are both more framework-aligned *and* more distinctive, while improvement patterns are harder to characterize specifically.

Task 3.1 vs. Task 3.2 Contrast A notable pattern emerges: our system performs relatively modestly on Task 3.1 (rank 11) but strongly on Task 3.2 (rank 2). This is consistent with the hypothesis that our approach optimizes for ABCD framework fidelity rather than lexical overlap with gold summaries. Task 3.1 is partly evaluated on ROUGE-L (lexical recall), which disadvantages summaries that correctly characterize the clinical dynamics but use different phrasing. Task 3.2 is evaluated on *Fit*, which directly rewards framework alignment, where our approach excels. This suggests that theory-explicit prompting is most beneficial for tasks requiring *conceptual* rather than *lexical* accuracy.

Deterioration vs. Improvement Asymmetry. Our strong deterioration performance relative to improvement suggests that maladaptive change follows more stereotyped ABCD patterns (self-criticism, hopelessness, and avoidance forming closed loops) that are robustly recurrent across individuals, while improvement signatures are more heterogeneous and harder to characterize across individuals. This asymmetry has implications for clinical monitoring: automated deterioration detection may generalize more reliably, while improvement characterization may require more individualized modeling.

4 Conclusion

We presented a theory-explicit prompting pipeline for longitudinal mental health modeling grounded in the MIND framework (Atzil-Slonim, 2025), encoding the ABCD taxonomy directly into the system prompt to produce clinically interpretable summaries and recurrent signatures. Our system ranked first in deterioration signature extraction and second overall in Task 3.2, achieving the highest Fit Score (1.000) across all teams—demonstrating that grounding NLP systems in established psychological theory is a performance-relevant design decision, particularly for tasks where conceptual accuracy matters more than lexical surface overlap.

The contrasting Task 3.1 performance highlights a broader tension between framework fidelity and lexical proximity to reference annotations, a challenge that remains open for clinically grounded generation.

Ethics Statement

This work uses a dataset of social media posts from mental health subreddits, handled in compliance with the CLPsych 2026 data access agreement. No individual users are identified, and all post examples in this paper have been paraphrased as required by the shared task guidelines. The systems described are research prototypes not intended for clinical deployment without human oversight; automated characterization of mental health states carries risks of misclassification with serious consequences for vulnerable individuals. All experiments use only open-source models, in compliance with the shared task requirement prohibiting the use of proprietary APIs.

Limitations

Our dataset of 40 Reddit timelines limits statistical reliability; the direction inference heuristic may mislabel ambiguous sequences, introducing noise into Task 3.2 aggregation, and the 90-word signature limit constrains the expressiveness of ABCD interaction descriptions. LLM outputs are non-deterministic; we report single inference passes without ensemble averaging. All results are specific to English-language Reddit posts from mental health subreddits and may not generalize to other languages, platforms, or clinical populations.

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A Appendix

A.1 Task Overview

The CLPsych 2026 Shared Task (Ali et al., 2026) focuses on longitudinal mental health modeling

within the MIND framework (Atzil-Slonim, 2025). This paper addresses the sequence-level tasks: structured summarization (Task 3.1) and recurrent signature extraction (Task 3.2), described below.

Task 3.1 – Sequence Summarisation. Given a chronologically ordered window of posts surrounding a detected change event (Switch or Escalation), systems generate a structured natural language summary describing how self-state dynamics evolve across the sequence and culminate in (for a Switch) or unfold through (for an Escalation) the change. Summaries describe: (1) the central recurring psychological theme in ABCD terms; (2) the dynamics within each self-state, including how ABCD subelements interact; and (3) the relationship between adaptive and maladaptive self-states, including patterns of dominance, suppression, or reflective dialogue. The direction of change (improvement or deterioration) must be stated explicitly (Ali et al., 2026).

Task 3.2 – Recurrent Signature Extraction. Building on Task 3.1, systems identify and summarise *recurrent* patterns of change across multiple timelines. Teams extract three signatures of deterioration and three signatures of improvement, each describing a recurring dynamic pattern in ABCD and self-state structure that leads up to and culminates in the change event. These signatures must be grounded in observable self-state dynamics across multiple individuals, rather than a single case (Ali et al., 2026).

A.2 Dataset

The shared task dataset comprises 40 Reddit-based user timelines drawn from mental health-related subreddits (MHS), split into 30 training timelines and 10 test timelines (Ali et al., 2026). User uniqueness is enforced so that no individual appears in both splits. Each timeline is a chronologically ordered sequence of posts annotated according to the MIND scheme (Atzil-Slonim, 2025).

For Tasks 3.1 and 3.2, sequence-level summaries around Switch and Escalation events are provided in the training split, released in stages over the course of the competition. Each summary is structured around the ABCD dynamics surrounding the change event, covering Central Theme, Within-State Dynamics, and Between-State Dynamics. The test data contain only post text and metadata; all annotations are withheld.